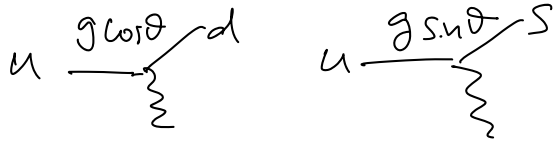
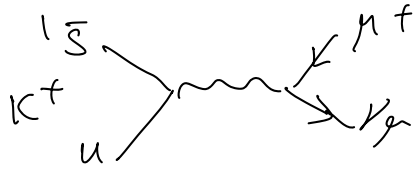


GIM mechanism

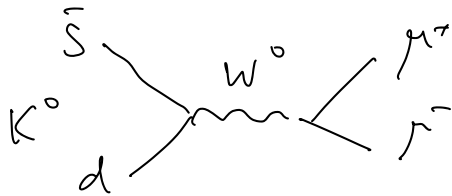
Cabibbo angle explains $\Gamma(\Delta S = 1) \ll \Gamma(\Delta S = 0)$
 weak interactions $u \rightarrow d' = \cos\theta \cdot d + \sin\theta \cdot s$



$$\Rightarrow \frac{\Gamma(\Delta S = 1)}{\Gamma(\Delta S = 0)} \approx \tan^2 \theta \approx \frac{1}{0.05} \quad \theta = 0.25$$



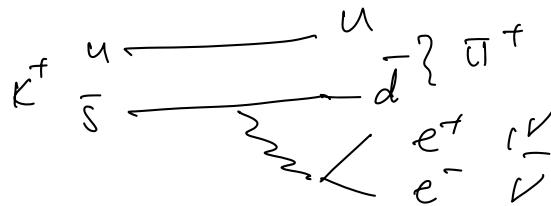
65%



10^{-9}

W^0 flavor changing neutral current.

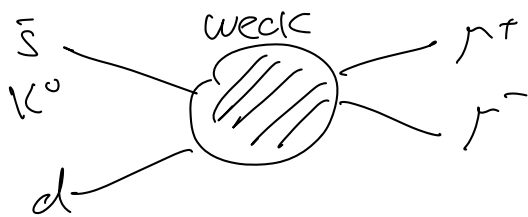
if W^0 exists



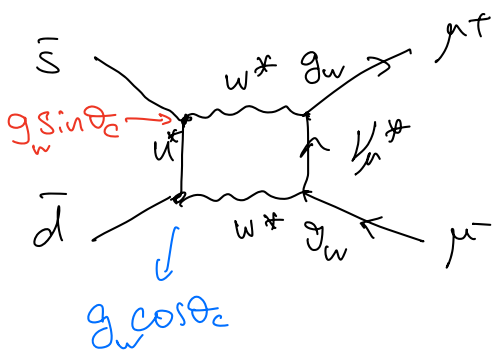
these are not seen.

How to explain $\Gamma(K^0 \rightarrow \pi^0 \pi^0) \approx 10^{-9} \ll 1$ but $\neq 0$

At leading order we cannot explain $\Gamma(K^0 \rightarrow \pi^0 \pi^0)$



at higher order.



$$\propto \frac{g_W \gamma^\mu}{q^2}$$

$$\propto \frac{g_W^2}{m_W^2} \frac{g_W^2}{m_W^2} \sin\theta_c \cos\theta_c$$

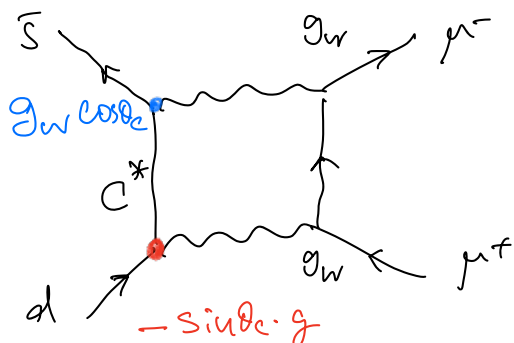
$$\Gamma(K^0 \rightarrow \pi^0 \pi^0) \propto G_F^4 \sin^2\theta_c \cos^2\theta_c \gg 10^{-9} \text{ (exp) }_{\text{mes}}$$

higher order predict $\neq$ measured rate.

solution proposed by Glashow-Iliopoulos-Maiani: (1970)

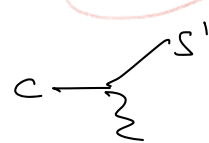
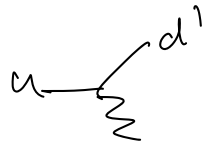
$\Rightarrow$ introduce a new up-like quark

$$Z_q = +\frac{2}{3}$$



$$u \rightarrow d' \quad d' = d \cdot \cos \theta_c + s \cdot \sin \theta_c$$

$$c \rightarrow s' \quad s' = -\sin \theta_c \cdot d + \cos \theta_c \cdot s$$



$$\mathcal{M}_C \propto \frac{g_w^2}{m_w^2} \frac{g_w^2}{m_w^2} (-\sin \theta_c)(\cos \theta_c)$$

$$\mathcal{M}(K^0 \rightarrow \pi^+ \pi^-) = \mathcal{M}_u(K^0 \rightarrow \pi^+ \pi^-) + \mathcal{M}_c(K^0 \rightarrow \pi^+ \pi^-)$$

$$= \left(\frac{g_w^2}{m_w^2} \right) \left(\frac{g_w^2}{m_w^2} \right) \left(\underbrace{\sin \theta_c \cos \theta_c \times \pi_u}_{\text{propagator } u} - \sin \theta_c \cos \theta_c \times \underbrace{(\pi_c)}_{\text{prop. } c} \right)$$

$$\pi_q = \frac{1}{q^2}$$

$\Gamma(K^0 \rightarrow \pi^+ \pi^-) \approx 10^{-9} \Rightarrow$ charm quark should have a mass of 1-3 GeV before discovery of J/psi

weak interactions.

$$\begin{pmatrix} u \\ d' \end{pmatrix} \begin{pmatrix} c \\ s' \end{pmatrix}$$

$$q_i \xrightarrow{g_w V_{ij}} q_j$$

$$W^{\pm}$$

$$V_{ij} = u, d, s, c$$

$$\begin{pmatrix} u & d \end{pmatrix} \begin{pmatrix} \cos \theta_c & \sin \theta_c \\ -\sin \theta_c & \cos \theta_c \end{pmatrix} \begin{pmatrix} c \\ s \end{pmatrix}$$

1 real parameter: Cabibbo angle

$$u \xrightarrow{g \cos \theta} d$$

$$\Delta S = 0$$

$$\Delta C = 0$$

$$u \xrightarrow{g \sin \theta} s$$

$$\Delta S = 1$$

$$c \xrightarrow{g \cos \theta} s$$

$$\Delta S = 1$$

$$\Delta C = 1$$

$$c \xrightarrow{-g \sin \theta} d$$

$$\Delta S = 0$$

$$\Delta C = 1$$

Flavor-changing charged weak current

$$\begin{pmatrix} u \\ d \end{pmatrix} \begin{pmatrix} c \\ s \end{pmatrix} \begin{pmatrix} t \\ b \end{pmatrix}$$

1973: Kobayashi-Maskawa
proposed 3rd family of quarks
to explain CP seen in 1964

$$q_i \xrightarrow{g_w V_{ij}} q_j$$

b: discovered 1977
t: discovered 1994

$$V_{ij} = V_{CKM} = \begin{pmatrix} V_{ud} = \cos\theta_c & V_{us} = \sin\theta_c & V_{ub} \\ V_{cd} = -\sin\theta_c & V_{cs} = \cos\theta_c & V_{cb} \\ V_{td} & V_{ts} & V_{tb} \end{pmatrix}$$

cabibbo
Kobayashi
Maskawa

V_{CKM} is a rotation matrix for field of quarks.

unitary matrix $V_{CKM}^\dagger V_{CKM} = \mathbb{1}$

3 real parameters (rotation angles) + 1 complex phase.

$$V_{CKM} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & c_{23} & s_{23} \\ 0 & -s_{23} & c_{23} \end{pmatrix} \begin{pmatrix} c_{13} & 0 & s_{13}e^{-i\delta} \\ 0 & 1 & 0 \\ -s_{13}e^{i\delta} & 0 & c_{13} \end{pmatrix} \begin{pmatrix} c_{12} & s_{12} & 0 \\ -s_{12} & c_{12} & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

cabibbo angle

$$= \begin{pmatrix} c_{12}c_{13} & s_{12}c_{13} & s_{13}e^{-i\delta} \\ -s_{12}c_{23} - c_{12}s_{23}s_{13}e^{i\delta} & c_{12}c_{23} - s_{12}s_{23}s_{13}e^{i\delta} & s_{23}c_{13} \\ s_{12}s_{23} - c_{12}c_{23}s_{13}e^{i\delta} & -c_{12}s_{23} - s_{12}c_{23}s_{13}e^{i\delta} & c_{23}c_{13} \end{pmatrix}$$

$$V_{CKM} = \begin{pmatrix} 1 - \lambda^2/2 & \lambda & A\lambda^3(\rho - i\eta) \\ -\lambda & 1 - \lambda^2/2 & A\lambda^2 \\ A\lambda^3(1 - \rho - i\eta) & -A\lambda^2 & 1 \end{pmatrix} + \mathcal{O}(\lambda^4)$$

Wolfenstein parameterization

$$\lambda = \sin\theta_c$$

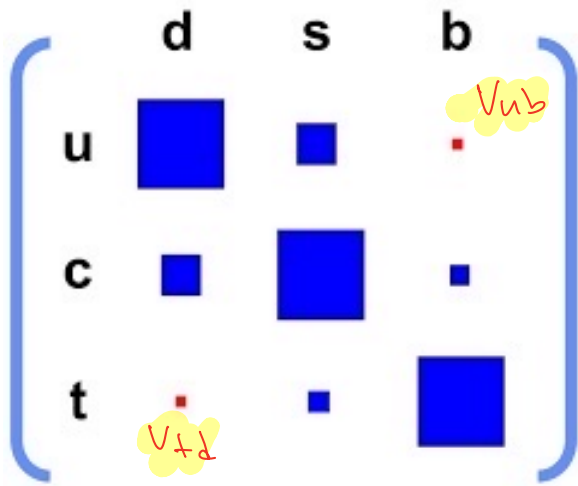
$$\theta_c \approx 0.25$$

$$1 - \frac{\lambda^2}{2} \approx 1$$

$$\text{Diagonals} \approx 1$$

$$\begin{pmatrix} u \\ d \end{pmatrix} \leftrightarrow \begin{pmatrix} c \\ s \end{pmatrix} \begin{pmatrix} t \\ b \end{pmatrix}$$

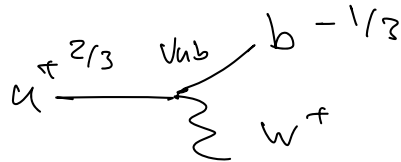
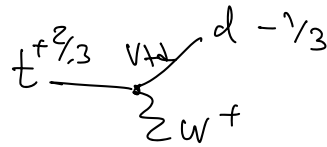
Smaller mat. elements.



Complex phase in V_{ub}, V_{td}

$$\text{area} \propto |V_{ij}|^2$$

$$\sin \theta_c = 0.25$$



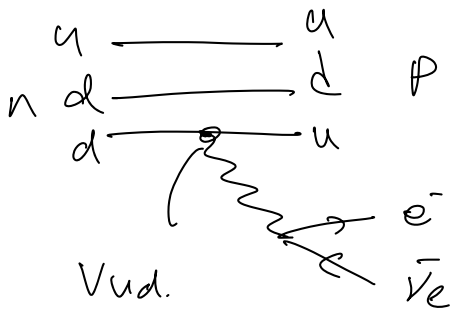
$$|V_{CKM}| = \begin{pmatrix} 0.97435 \pm 0.00016 & 0.22500 \pm 0.00067 & 0.00369 \pm 0.00011 \\ 0.22486 \pm 0.00067 & 0.97349 \pm 0.00016 & 0.04182^{+0.00085}_{-0.00074} \\ 0.00857^{+0.00020}_{-0.00018} & 0.04110^{+0.00083}_{-0.00072} & 0.999118^{+0.000031}_{-0.000036} \end{pmatrix}$$

$$V_{CKM}^\dagger V_{CKM} = \mathbb{1}$$

$$\Gamma(u \rightarrow p e \bar{\nu}_e) \propto |V_{ud}|^2$$

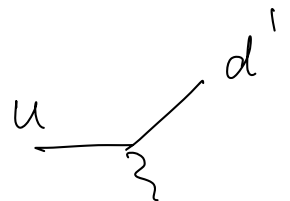
$$\Gamma(u \rightarrow p e \bar{\nu}_e) = \text{BF}(u \rightarrow p e \bar{\nu}_e) \times \Gamma_{\text{tot}}$$

$$\frac{\#(u \rightarrow p e \bar{\nu}_e)}{\# \text{ neutrons}} \Rightarrow \text{exp. meas. of } |V_{ud}|$$



Int. double

$$\bar{\psi}_{d'} \gamma^\mu (1 - \gamma^5) \psi_u$$



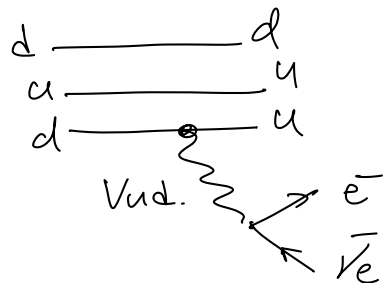
$$\bar{\psi}_d(V) \gamma^\mu (1 - \gamma^5) \psi_u$$

To measure V_{ij}

$\Gamma(A \rightarrow B) \propto |V_{ij}|^2$

need hadrons with q_i or q_j

V_{ud}

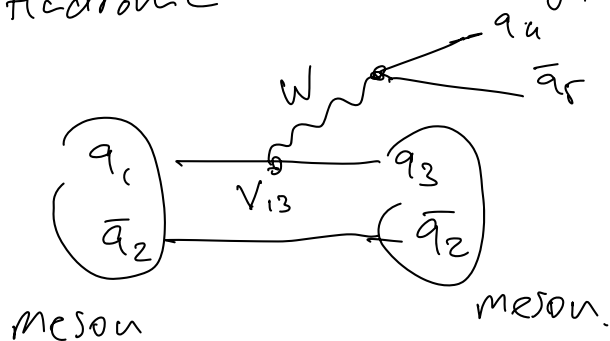


V_{ud}	V_{us}	V_{ub}
$\pi \rightarrow \ell \nu$	$K \rightarrow \ell \nu$	$B \rightarrow \pi \ell \nu$
	$K \rightarrow \pi \ell \nu$	
V_{cd}	V_{cs}	V_{cb}
$D \rightarrow \ell \nu$	$D_s \rightarrow \ell \nu$	$B \rightarrow D \ell \nu$
$D \rightarrow \pi \ell \nu$	$D \rightarrow K \ell \nu$	$B \rightarrow D^* \ell \nu$
V_{td}	V_{ts}	V_{tb}
$B_d \leftrightarrow \bar{B}_d$	$B_s \leftrightarrow \bar{B}_s$	

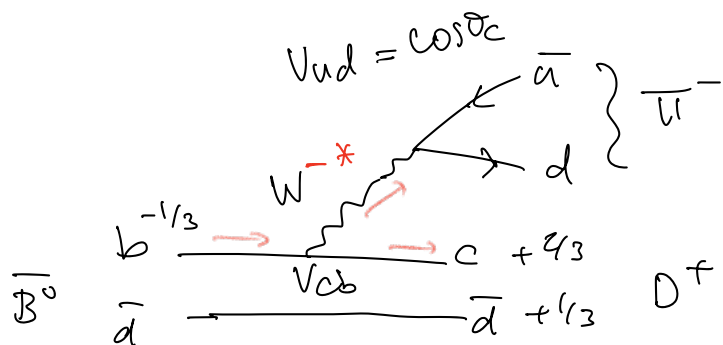
$\bar{B}$ decay $\propto n \rightarrow p e^- \bar{\nu}_e$

Hadronic weak decay.

$\Rightarrow$ involve 2 CKM matrix elements.

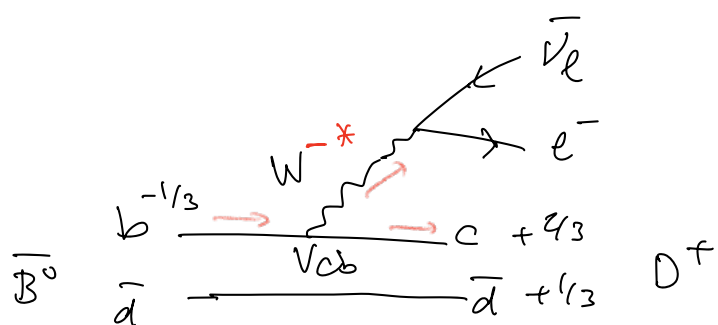


$$\mu \sim |V_{13}|^2 |V_{45}|^2$$



$$BF(B^0 \rightarrow D^+ \pi^-) \propto |V_{ud}|^2 |V_{cb}|^2$$

Generally semileptonic and leptonic decays preferred.



$$M_B = 5.279 \text{ GeV}$$

$$M_D = 1.86 \text{ GeV}$$

$$\bar{B}^0 \rightarrow D^+ e^- \bar{\nu}_e$$

$$Q = m_B - m_D - m_\ell$$

$$m_\tau = 1.77 \text{ GeV}$$

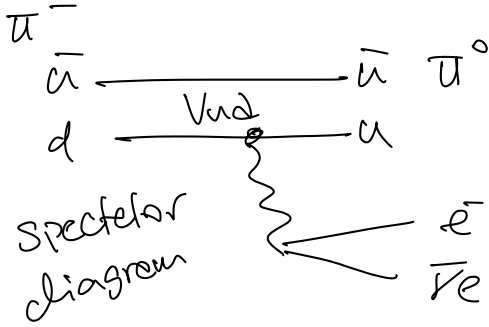
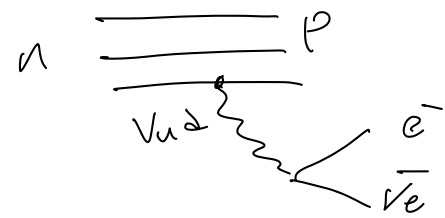
$$D^+ \mu^- \bar{\nu}_\mu$$

$$D^+ \tau^- \bar{\nu}_\tau$$

$$|M|^2 |V_{cd}|^2$$

Prefer mesons to baryons.

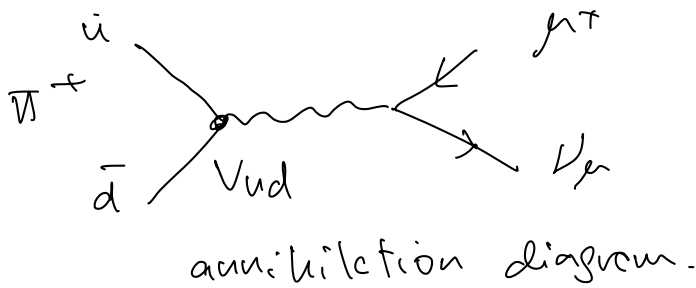
$$\Gamma \propto |V_{ud}|^2 \times (\text{QCD Corrections})$$



$$\pi^- \rightarrow \pi^0 e^- + \bar{\nu}_e$$

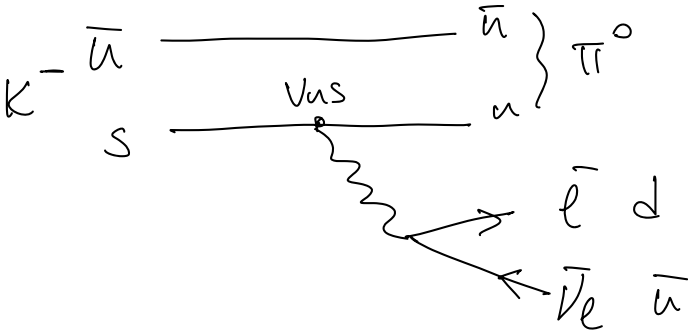
$$m_{\pi^-} > m_{\pi^0}$$

$$\Gamma \propto |V_{ud}|^2 \times (\text{QCD Corrections})$$



$$\Gamma \propto |V_{ud}|^2$$

V_{us}

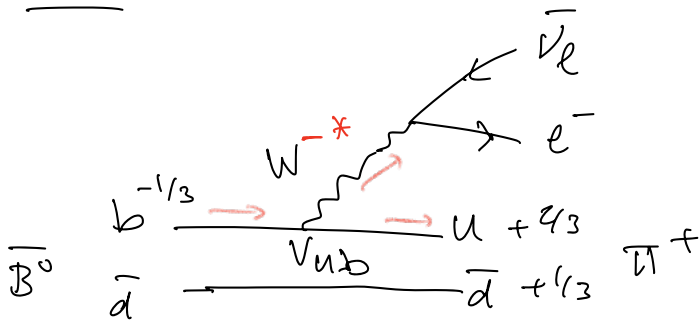


$$K^+ \rightarrow \pi^0 \ell^+ \nu_\ell \propto |V_{us}|^2$$

$$\pi^0 \pi^+ \propto |V_{us}|^2 |V_{ud}|^2$$

$$K^+ \rightarrow \mu^+ \nu_\mu \pi^0$$

V_{ub}



$$\text{BF}(B \rightarrow \pi \ell \nu) \propto |V_{ub}|^2$$

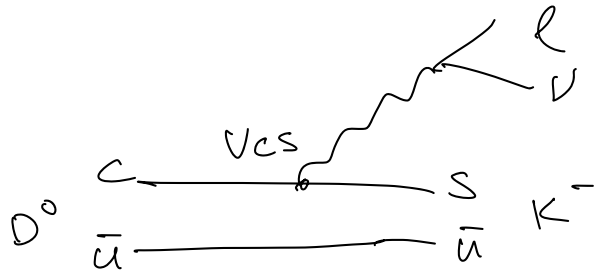
K⁺ DECAY MODES	Fraction (Γ_i/Γ)	Scale factor/ Confidence level (MeV/c)	p
Leptonic and semileptonic modes			
$e^+ \nu_e$	$(1.582 \pm 0.007) \times 10^{-5}$		247
$\mu^+ \nu_\mu$	$(63.56 \pm 0.11) \%$	$S=1.2$	236
$\pi^0 e^+ \nu_e$	$(5.07 \pm 0.04) \%$	$S=2.1$	228
Called K_{e3}^+			
$\pi^0 \mu^+ \nu_\mu$	$(3.352 \pm 0.033) \%$	$S=1.9$	215
Called $K_{\mu 3}^+$			
$\pi^0 \pi^0 e^+ \nu_e$	$(2.55 \pm 0.04) \times 10^{-5}$	$S=1.1$	206
$\pi^+ \pi^- e^+ \nu_e$	$(4.247 \pm 0.024) \times 10^{-5}$		203
$\pi^+ \pi^- \mu^+ \nu_\mu$	$(1.4 \pm 0.9) \times 10^{-5}$		151
$\pi^0 \pi^0 \pi^0 e^+ \nu_e$	$< 3.5 \times 10^{-6}$	$CL=90\%$	135
Hadronic modes			
$\pi^+ \pi^0$	$(20.67 \pm 0.08) \%$	$S=1.2$	205
$\pi^+ \pi^0 \pi^0$	$(1.760 \pm 0.023) \%$	$S=1.1$	133
$\pi^+ \pi^+ \pi^-$	$(5.583 \pm 0.024) \%$		125

Charm mesons: $D^\pm, \bar{D}^0, D^0 \Rightarrow V_{cd}, V_{cs}$

$$D \rightarrow K \ell \nu \propto |V_{cs}|^2$$

$$D \rightarrow \pi \ell \nu \propto |V_{cd}|^2$$

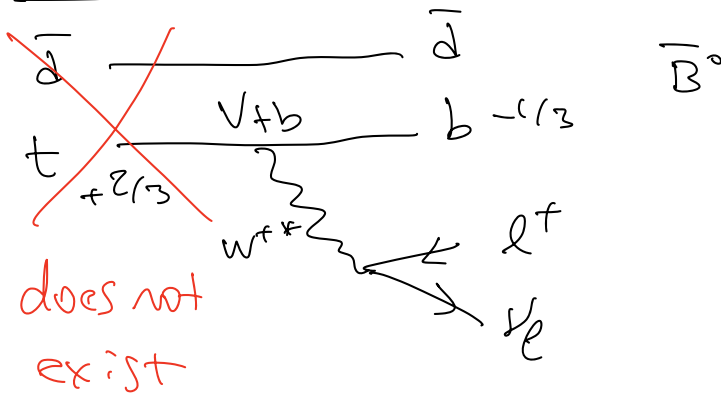
$$m_D = 1.864 \text{ GeV}$$



$$\frac{|V_{cd}|^2}{|V_{cs}|^2} \simeq 0.05 \ll \frac{\rho(D \rightarrow K)}{\rho(D \rightarrow \pi)}$$

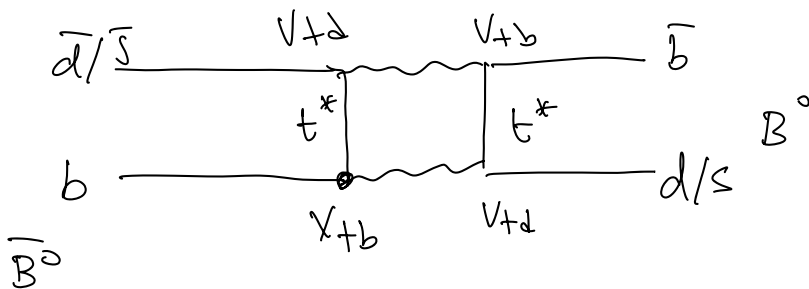
CKM wins over phase space.

V_{tx}



$\Rightarrow$ measurement of V_{tx} requires top in loops of higher order.

$|V_{tb}| \simeq 1 \Rightarrow$ look B mesons.



$$\mu \sim |V_{tb}|^2 |V_{td}|^2$$

$$m_{B^0} \equiv m_{\bar{B}^0}$$

$$\bar{B}^0 \leftrightarrow B^0 \quad \Delta b = 2$$

$$\bar{B}_s^0 \leftrightarrow B_s^0$$

neutral meson oscillation.

$$K^0 \leftrightarrow \bar{K}^0$$

$$D^0 \leftrightarrow \bar{D}^0$$